

Managing Water-Energy-Carbon Nexus for Urban Areas With Ambiguous Moment Information

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Abstract—The rapid growth of electric load fuels urbanization but creates unsettling environmental challenges. In that regard, the energy system is a crucial part of consuming water and emitting greenhouse gases. To support reliable and economical energy system operations, with reduced water waste and carbon emissions, we design a viable co-optimization of water-energy-carbon nexus for an integrated energy system by adequately modelling essential interdependencies of power, gas, and water systems. A two-stage distributionally robust optimization method is proposed: the first-stage model minimizes the day-ahead reserve capacity scheduling for the next day, and the second-stage model enables real-time dispatch by considering the variability in renewable power generation. The carbon emissions by power generation and distribution are incorporated in both stages to ensure low-carbon operations. We use a two-stage moment-based distributionally robust approach to capture and represent the uncertainties in renewable generation. Instead of assuming a deterministic distribution, we characterize an ambiguity set with mean vectors and covariance matrices, which produces a family set of distributions. We conduct case studies using a modified IEEE 33-bus power system that includes and links a 20-node gas system and a 10-node water system. The comparative results indicate that the proposed co-optimization of water-energy-carbon nexus is superior than several benchmarks. By considering the close interdependencies of water, energy, and carbon, we create an optimal resource allocation scheme, which is capable of minimizing operation costs, lessening carbon emissions, and reducing water waste. The current research enables effective operation schemes for urban energy systems, illustrates a viable

way to curb local environmental emissions, and increases water usage efficiency in the water-energy-carbon nexus.

Index Terms—Carbon reduction, energy management, renewable energy sources, water-energy-carbon nexus system.

NOMENCLATURE

A. Sets

t	Set of time slots.
b	Set of power buses.
n, w	Set of gas and water nodes.
i_e	Set of conventional distributed generators (DG).
i_g	Set of natural gas sources.
wr, wp	Sets of water reservoirs and water pumps.
j	Set of renewable DGs.
gt	Set of gas turbines.
l_e, l_g	Sets of power lines and gas pipelines.
l_{wp}, l_w	Sets of water pipelines with/without pumps.
k_e, k_g, k_w	Sets of power, gas, and water loads.

B. Parameters

$P_{k_e, t}, Q_{k_e, t}$	Load of active and reactive power.
$P_{m, \max}$	Highest power purchase from upper level market.
$R_{i_e}^+, R_{i_e}^-, R_{gt}^+, R_{gt}^-, R_{wp}^+, R_{wp}^-$	Limits of up, down reserve capacity of conventional DGs, the gas turbine and water pumps.
$P_{i_e, \max}, P_{i_e, \min}, P_{gt, \max}, P_{gt, \min}, P_{wp, \max}, P_{wp, \min}$	Limits of active power output of conventional DGs, gas turbine output and water pump power consumption.
$Q_{i_e, \max}, Q_{i_e, \min}$	Limits of DG reactive power output.
$V_{b, \max}, V_{b, \min}$	Voltage limits.
x_{l_e}, r_{l_e}	Reactance and resistance of power lines.
V_0	Reference voltage magnitude.
$f_{l_e, \max}, qf_{l_e, \max}$	Limits of active and reactive power flow.
$\omega_j^s(t)$	Forecasted output of renewable DG.
$G_{k_g, t}$	Gas load.
$G_{i_g, \max}, G_{i_g, \min}$	Limits of gas source output.
$Pr_{l_g, \max}, Pr_{l_g, \min}$	Limits of gas pressure of gas pipelines.
γ_{l_g}	Weymouth equation constant.
$f_{l_g, \max}$	Maximum gas flow.
CF_{l_g}	Gas compressor coefficient.
η_e	Electrical efficiency for electrolyser.

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$P_{wr,max}$	Limit of water purchase from reservoir.
$P_{k_w,t}$	Water load.
$h_{w,max}^{l_{wp}}, h_{w,min}^{l_{wp}}$	Limits for head pressure of water node connected with/without water pump.
$a_{l_{wp}}, b_{l_{wp}}$	Water pump characteristic coefficients.
$R_{l_{wp}}, R_{l_w}$	Head gain and loss coefficients.
π_{wp}	Water pump efficiency.
$f_{l_{wp},max}^s, f_{l_w,max}^s$	Water flow for water pipeline with/without pumps.
c_{i_e}, c_{i_g}, c_{wp}	Carbon emission coefficients of generators.
c_{l_e}, c_{l_g}	Carbon emission coefficients of branches.
$\lambda_{i_e}^a, \lambda_{i_e}^b, \lambda_{i_e}^c$	Cost coefficients of conventional DG output.
λ_{i_g}	Cost coefficient of natural gas source output.
λ_m, λ_{wr}	Cost coefficient of power and water purchase.
$\lambda_{i_e}^+, \lambda_{i_e}^-, \lambda_{GT}^+, \lambda_{GT}^-, \lambda_{wp}^+, \lambda_{wp}^-$	Cost coefficients for up and down reserve of conventional DGs, gas turbines, and water pumps.
$\lambda_{\{\cdot\}}^{re}$	Adjustment cost coefficient.

C. Variables

$P_{m,t}, P_{wr,t}$	Active power purchase.
$r_{i_e,t}^+, r_{i_e,t}^-, r_{gt,t}^+, r_{gt,t}^-, r_{wp,t}^+, r_{wp,t}^-$	Up and down reserve capacity of conventional DGs, the gas turbine, and water pumps.
$P_{i_e,t}, P_{gt,t}, P_{wp,t}$	Active power output of conventional DGs, gas turbine output, and water pump power consumption.
$Q_{i_e,t}$	Reactive power output of conventional DGs.
$V_{b,t}^s, V_{b,t}^{re}$	Scheduled and adjusted voltage of bus b at time t.
$f_{l_e,t}, qf_{l_e,t}, f_{l_g,t}$	Active and reactive power flow and gas flow.
$G_{i_g,t}$	Gas source output.
$f_{l_g,t}$	Gas flow.
$P_{gt,t}$	Gas turbine output.
$Pr_{n,t}$	Pressure of gas node n.
$f_{l_g,GT,t}, P_{gt,t}$	Gas flow injection and output of gas turbine.
$PP_{n,t}^{2G}$	Power consumed by the electrolyser.
$P_{wr,t}$	Water purchase.
$h_{w,t}^{l_{wp}}, h_{w,t}^{l_w}$	Water pressure of pipe with/without water pumps.
$P_{wp,t}$	Water pump power consumption.
$\bar{h}_{w,t}^{l_w}, \bar{h}_{w,t}^{l_{wp}}$	Elevation of water node linked with/without pumps.
$\tilde{h}_{w,t}^{l_w}, \tilde{h}_{w,t}^{l_{wp}}$	Head loss and gain of water node.
$f_{l_{wp},t}, f_{l_w,t}$	Water flow of pipe with/without water pumps.

I. INTRODUCTION

WATER and energy are fundamentally crucial to urbanization and economic developments [1]. However, stressing water and energy issues create urgent challenges in modern society and humanity [2]. For instance, 70% of the global population are expected to face severe water shortage crises by 2050, according to the estimated water shortage trend [3]. Meanwhile, energy consumption has been increasing at an alarming rate that substantially outpaces that of freshwater production [4]. In many urban management systems, water and energy systems are designed to operate independently, without considering their interrelationships, linkages, and couplings [5]. These two systems are intrinsically related and highly interdependent [6]. Take an ordinary city for illustration, water systems, which contribute about 3% of the power supply, usually consume a large proportion of its total energy [7]. According to the United States (U.S.) power statistics [8], in urban areas, water pumping and distributions consume nearly 80% of the power consumption in the water sector. Similarly, energy systems account for a large proportion of the total water demands.

The water-energy nexus entails an efficient utilization of the water and energy infrastructures in a holistic framework. Research efforts have been undertaken to study the joint optimization of water and power systems, with a common focus on operation cost reduction and system efficiency improvement. For example, Oikonomou and Parvania [9] suggest a co-optimization model for optimal water-power flow in a distribution system, formulating the dynamic operation of water treatment and desalination plants as a mixed-integer linear programming model. Li et al. [10] devise an optimal demand response program for the micro water-energy nexus at a distribution level. In the proposed program, water systems are utilized to manage renewable energy sources. Mehrjerdi [11] develops a hybrid water-energy supply system for remote islands and employs various desalination procedures in an optimization model. Wang et al. [12] suggest a robust optimization (RO)-based operation model for an integrated water-energy system, with the consideration of wind uncertainty. A two-step solution procedure using a mixed-integer second-order cone program is proposed to address the difficulty that result from the interdependencies among heterogeneous energy systems. Moazeni et al. [13] develop a co-management framework for an islanded water-energy system to minimize electricity costs. Three operation scenarios are considered: standalone, grid-connected, and storage-equipped operations. To improve the operational efficiency of smart islands, Mohamed et al. [14] design a water-energy microgrid integrated with smart transportation systems. Bilateral flow connections allow seawater supplements between the secondary tank and the water grid. Zuloaga and Vittal [15] simulate long-term optimization decisions that involve both drought effects and power outages. Different control policies are implemented to explore the benefits of nexus management for greater operational efficiency and resilience. Another stream of water-energy nexus management research targets optimization for small-scale systems [16], [17]. For example, Pezhmani et al. [16] apply a centralized stochastic

optimization (SO) for networked water-energy microgrids, with the consideration of trading behaviours. Tabar et al. [17] develop an optimal operation scheme for renewable energy-water nexus at the hub level and use a scenario-based SO to handle wind uncertainties.

Carbon emissions created by fuel combustion, energy transmission and distribution, water pumping, and waste water treatment represent a severe challenge in global climate [18]. In the U.S., 290 million metric tons of carbon emissions are attributed to water distribution, treatment, and heating [19]. The power generation in the U.S. creates 2.06 billion tons of carbon dioxide in 2018 [20]. In addition to the water-energy nexus, the water-energy-carbon nexus is suggested and has attracted a growing attention for environmental sustainability [21]. Most analyses of the water-energy-carbon nexus emphasize input-output model designs [22], [23], [24], [25], [26]. For example, Wang et al. [23] suggest an environmental input-output model-based method to estimate total water and energy consumption, as well as carbon emissions, for assessing the water-energy-carbon nexus in different sectors in China. Heavy, light, and service industries appear to be three top sectors that create the highest indirect energy consumption and carbon emissions. Yang et al. [24] employ a combination of the direct resource consumption and emissions in an environmental input-output model on an urban scale, according to monetary flows. They compare different sectors and examine the sustainability of Beijing and Shanghai. Venkatesh et al. [25] analyse and compare the metabolism of urban water systems in four cities, and report that water-energy-carbon nexus analyses help water agencies to effectively assess energy intensity and greenhouse gas emissions for alternative planning options.

Carbon emission modelling is central to the water-energy-carbon nexus. Previous research provides valuable insights into carbon emission modelling for energy systems. For example, Kang et al. [27], [28] rely on the optimal carbon emission flow concept for transferring emission responsibilities from the producer to consumer side. Yang et al. [29] employ a combined power-to-gas (P2G) and gas-fired power plant system to reduce carbon emissions, which considers power-gas-heat flow. Akbari-Dibavar et al. [30] design a low-carbon, economical emission optimization model for a combination of renewable and coal power plants. Partial carbon emissions are decreased through a carbon capture system. Cheng et al. [31] develop a bi-level expansion planning model for integrated energy systems (IESs), in which carbon emission flows are employed for emission cap allocation. To reduce energy customers' reliance on producers of high emission intensity, Wang et al. [32] propose a low-carbon demand response model that combines energy operations and demand, while considering carbon intensity controls. Algarni et al. [33] study the role of emission rate-based carbon tax for emission reduction in multi-area power systems.

There is an increasing need for effective integration of multi-energy vectors and traditional power systems, due to the energy decarbonization trend and advanced coupling technologies for energy infrastructures. An IES involves various, complementary energy conversion and generation facilities; it is designed to harness increased energy efficiency, facilitate renewable

integration, and meet environmental challenges and global climate issues. Extensive research has been conducted to optimize IES operations, in terms of economic efficiency and effectiveness for environmental issues. However, the volatility and variability of renewable generation often constrain the economic optimality and system reliability of an IES significantly. More previous research relies on stochastic optimization (SO) and robust optimization (RO) for addressing the renewable generation uncertainties [34], [35], [36], [37]. For example, Ceseña and Mancarella [36] develop a robust operational optimization framework and uses a real smart district in the United Kingdom to demonstrate its effectiveness. In this framework, RO is applied to capture the complexity associated with uncertainties and inter-temporal constraints. Also, Mei et al. [37] design stochastic optimal operations for an IES in an industrial park, and analyse stochastic scenarios using Latin hypercube sampling and K-means scenario reduction.

In general, SO addresses uncertainties via sampling scenarios; typically, it assumes that the uncertainty distribution is perfectly known and then uses it to generate uncertainty samples, according to the (assumed) deterministic distribution information. The practicality and value of SO have been criticized, because it is difficult to obtain accurate distribution information in many real-world scenarios. Additionally, SO cannot provide reliable solutions without sufficiently large data sets. As a partial remedy, RO requires modest uncertainty data (e.g., mean, range information) and avoids uncertainty estimations using a predefined uncertainty set. However, RO often yields sub-optimal solutions, due to its focus on the worst-case perturbation with excessive conservativeness.

Distributionally robust optimization (DRO) offers a promising optimization approach to bridge SO and RO for leveraging the strengths of each. It characterizes uncertainties with an ambiguity set that can be constructed with essential statistical information (e.g., moment information, discrepancy range). By and large, DRO mitigates SO's overfitting behaviours that stem from the deterministic distribution condition, and does not require explicit knowledge about the uncertainty distribution that is assumed (required) by SO. Moreover, its worst-expectation setting can mitigate the excessive conservativeness of RO. Previous research has applied DRO to address system uncertainties in energy operations. For example, Lu et al. [38] develop an optimal management scheme for energy distribution companies that can utilize electric vehicle aggregators flexibly. This method is also employed to evaluate the operation cost associated with renewable generation uncertainties and EVA charging loads. Liu et al. [39] study a real-time economic dispatch and formulate an automatic generation control in which DRO is applied to model the uncertainties in energy load and renewable generation. Zhou et al. [40] suggest resilient operations for multi-energy systems to hedge against extreme weather, and depend on Wasserstein metric for characterizing uncertain contingencies.

This study proposes a two-stage co-optimization method for the water-energy-carbon nexus in a low-carbon water-energy system (LCWES), which includes power, gas and water systems. The extensive interdependencies among these distinct systems

are modelled explicitly and adequately. The operational efficiency of LCWES, which considers the close linkages of water, energy, and carbon, is realized through an optimal allocation of water and energy resources, while minimizing carbon emissions. Carbon emissions by power, water and gas systems are also modelled, with a consideration of the related charges for the overall operation cost. A two-stage DRO model is developed to capture and represent the uncertainties in renewable generation. It includes day-ahead and real-time operation schemes, before and after the uncertainty effectuation. Instead of assuming or obtaining a deterministic uncertainty distribution, we model the uncertain renewable generation with a more realistic and reliable moment ambiguity set. As a result, the original problem can be dualized and is then reformulated into a tractable semidefinite programming problem. We design a constraint generation-based relaxation iterative method (CG-RIM) to solve the problem iteratively. The proposed LCWES management (LCM) provides system operators with a two-stage operation scheme to minimize operation costs, increase overall system efficiency, and reduce carbon emissions, while dealing with enormous cross-vector interdependencies. The evaluation results are valuable for facility or system operators who are equipped with detailed analyses and in-depth understanding of the complex water-energy-carbon nexus.

The key contribution of this paper is summarized as following: we develop a coordinated operation model for water-energy systems that consider carbon emissions. The intricate nexus among power, gas, water and carbon is modelled explicitly and adequately. Our LCM model can reduce carbon emissions in energy generation, distribution and conversion, and thereby support the grand net-zero emission goal. The related carbon emissions generated in the LCM are considered and the emission penalty charges are incorporated in the overall optimization objective. As a result, system metabolism is reduced and sustainability is improved.

Second, the proposed two-stage DRO model is valuable and practical, because it is capable of solving key issues in the water-energy-carbon nexus problem, without large, ambiguous data sets. Compared with a single-stage optimization model for one-time decision making, our two-stage model provides additional operational flexibility, i.e., day-ahead preparatory actions and real-time corrective measures. Moreover, it overcomes the limitations of SO and RO by leveraging partial distributional information with moderate robustness. With this method, we can construct a moment-based ambiguity set to accommodate a set of uncertainty distributions, reformulate the original problem into a tractable semidefinite programming problem, and then apply a constraint generation algorithm to solve the reformulated complex LCM problem.

The remainder of this paper is organized as follows. Section II presents the objective function and constraints for both day-ahead and real-time stages. Section III describes the proposed method for solving the two-stage DRO model. Section IV details our case studies and reports key results that demonstrate the greater utilities and values of LCWES in comparison with several benchmarks. Finally, we conclude the paper in Section V.

II. PROBLEM FORMULATION

The proposed LCWES contains power, gas and water systems with related carbon emissions. The illustrative description of the proposed water-energy-carbon nexus is given in Fig. 1, where the enormous interdependencies among each system are achieved by energy converters mutual reliance. The power from the power system realizes P2G conversion and supplies water pumps. The water system supplies P2G facilities during the electrolysis process and is also required for natural gas sources. Gas turbines with power conversion enable to supply the power system as well as water pumps. Meanwhile, inevitably carbon metabolism is released. As shown in the outputs of the proposed water-energy-carbon nexus, the proposed LCM can achieve improved energy efficiency, optimal water utilization, economic effectiveness, and reduced emissions. Notably, we propose an uncertainty-aware optimization model for the IES considering the coordination between power and gas systems by optimally and flexibly dispatching energy converters with the maximal usage of renewable energy sources. Meanwhile, the optimization for the water system considers a complete series of technical constraints, aiming to provide a wise and sustainable water utilization scheme and meanwhile satisfy the water requirement of energy systems.

The proposed two-stage optimization framework of LCWES is given in sections A and B, followed by the description of the objective function in section C. Note that it is assumed that the proposed water-energy-carbon nexus system is owned and controlled by a single entity. It is to be noted that the superscript ‘s’ and ‘re’ represent the variables in the first and second stages, respectively.

A. Day-Ahead Operation of LCM

The first-stage day-ahead operation corresponds to the current decision-making before the realization of the renewable generation uncertainties. It schedules the generation plan of power, gas, and water systems considering the technical constraints of energy and water infrastructures. (1)–(33) are the first-stage constraints. Constraint (1) shows the power purchase $P_{m,t}^s$ limit from the upper-level market. Constraints (2) and (3) adjust the up and down reserve capacity of generators ($r_{\{ \},t}^+$ and $r_{\{ \},t}^-$). The indices of traditional DGs, gas turbines and water pumps are represented by the indices ie , gt , and wp . Constraints (4) and (5) limit the output of traditional DGs, gas turbines and water pumps ($P_{ie,t}^s$, $P_{gt,t}^s$ and $P_{wp,t}^s$). The reactive power of traditional DGs $Q_{ie,t}^s$ is adjusted in (6). (7) is the constraint of voltage magnitude $V_{b,t}^s$ at bus b . The relationship between the power flow and bus voltage is described in (8) and (9) via linearized DistFlow equation. The active and reactive power flow are denoted as $f_{l_e,t}^s$ and $qf_{l_e,t}^s$. x_{l_e} and r_{l_e} represent the reactance and resistance of power line l_e . When the power flow is injected into a power line, the voltage at the initial and terminal buses are represented by $V_b^{s,ini}$ and $V_b^{s,ter}$, respectively. Constraints (10) and (11) ensure the balance of active and reactive power, where $\omega_{j,t}^s$ is the renewable forecast generation; $f_{l_e,t}^{s,ini}$ and $f_{l_e,t}^{s,ter}$ are the active power flow at the initial and terminal buses; $P_{gt,t}^s$ and

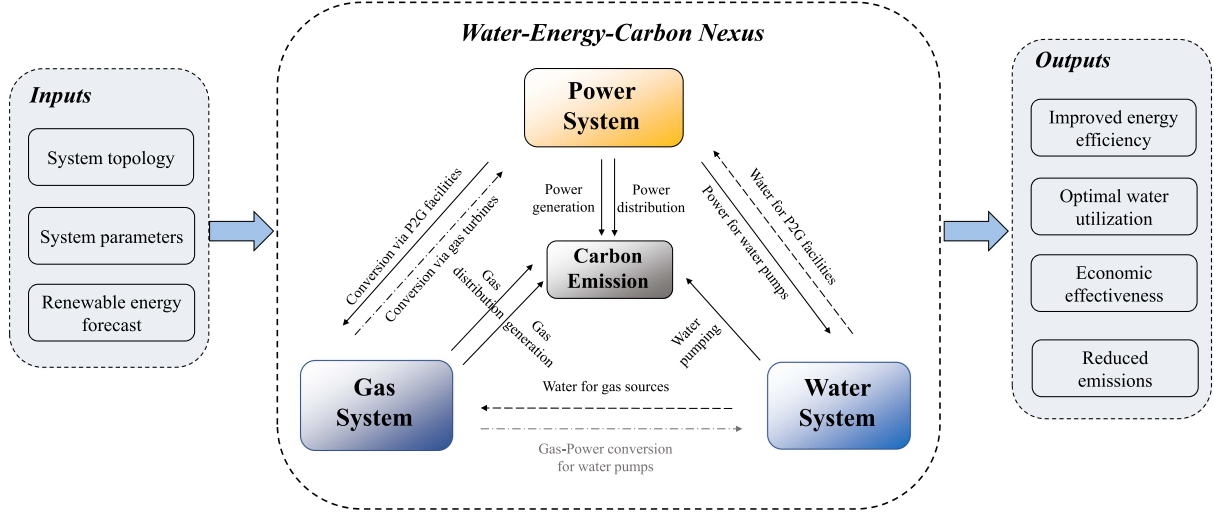


Fig. 1. The proposed water-energy-carbon nexus.

$P_{n,t}^{s,P2G}$ are the external injection from gas turbines and P2Gs; $P_{wp,t}^s$ is the power consumption of the water pump; $P_{ke,t}$ is the power load demand.

$$0 \leq P_{m,t}^s \leq P_{m,\max}$$

$$0 \leq r_{\{\cdot\},t}^+ \leq R_{\{\cdot\}}^+, \{\cdot\} = i_e, gt, wp$$

$$0 \leq r_{\{\cdot\},t}^- \leq R_{\{\cdot\}}^-, \{\cdot\} = i_e, gt, wp$$

$$P_{\{\cdot\},t}^s + r_{\{\cdot\},t}^+ \leq P_{\{\cdot\},\max}, \{\cdot\} = i_e, gt, wp$$

$$P_{\{\cdot\},\min} \leq P_{\{\cdot\},t}^s - r_{\{\cdot\},t}^-, \{\cdot\} = i_e, gt, wp$$

$$Q_{ie,\min} \leq Q_{ie,t}^s \leq Q_{ie,\max}$$

$$V_{b,\min} \leq V_{b,t}^s \leq V_{b,\max}$$

$$V_b^{s,ini} - V_b^{s,ter} = (f_{l_e,t}^s r_{l_e} + q f_{l_e,t}^s x_{l_e}) / V_0$$

$$0 \leq \{\cdot\}_{l_e,t}^s \leq \{\cdot\}_{l_e,\max}^s, \{\cdot\} = f, qf$$

$$\begin{aligned} & \sum_{ie \in I_e} P_{ie,t}^s + \sum_{j \in J} \omega_{j,t}^s + \sum_{l_e \in L_e} f_{l_e,t}^{s,ini} - \sum_{l_e \in L_e} f_{l_e,t}^{s,ter} + P_{gt,t}^s \\ &= \sum_{ke \in K_e} P_{ke,t} + \sum_{l_e \in L_e} f_{l_e,t}^{s,inj} + \sum_{n \in N} P_{n,t}^{s,P2G} + \sum_{wp \in WP} P_{wp,t}^s \end{aligned} \quad (10)$$

$$\sum_{ie \in I_e} Q_{ie,t}^s + \sum_{l_e \in L_e} q f_{l_e,t}^{s,ini} - \sum_{l_e \in L_e} q f_{l_e,t}^{s,ter} = \sum_{ke \in K_e} Q_{ke,t} \quad (11)$$

Equation (12) depicts the output constraint of natural gas sources, where $G_{ig,t}^s$ is the gas output, limited by the upper and lower output $G_{ig,\max}$ and $G_{ig,\min}$. Constraint (13) limits the pressure $Pr_{l_g,t}^s$ at gas nodes by the upper and lower bounds $Pr_{l_g,\max}$ and $Pr_{l_g,\min}$. Constraint (14) guarantees the unidirectional gas flow. In (15), the unidirectional Weymouth gas flow equation is applied to describe the relationship between gas flow and node pressure, where γ_{l_g} represents the Weymouth equation

coefficient. Weymouth gas flow equation was developed to simulate compressible gas flow in long gas pipelines. Thanks to the fact that the topology of urban gas systems is usually radial, we degrade the original left-hand side $|f_{l_g,t}^s| f_{l_g,t}^s$ to $f_{l_g,t}^{s2}$ in (15) [41], [42]. (16) constraints the gas flow under the maximum magnitude $f_{l_g,\max}^s$. In (17), the power output of gas turbines $P_{gt,t}^s$ is given, where c_{GT} and $f_{l_g,gt,t}^s$ are the conversion efficiency and gas injection. Constraint (18) characterizes the P2G conversion via electrolysis process, where η_e is the electrical efficiency of electrolyser; Ω_{hy} is the gross calorific value for hydrogen; $P_{n,t}^{s,P2G}$ and $G_{n,t}^{s,hy}$ are the input and output of the P2G facility, respectively. Finally, the balancing condition of gas system is presented in (19), where $f_{l_g,t}^{s,ini}$ and $f_{l_g,t}^{s,ter}$ are the gas flow at the initial and terminal nodes; $G_{k_g,t}$ is the gas load demand.

$$G_{ig,\min} \leq G_{ig,t}^s \leq G_{ig,\max} \quad (12)$$

$$Pr_{l_g,\min}^2 \leq Pr_{l_g,t}^{s2} \leq Pr_{l_g,\max}^2 \quad (13)$$

$$Pr_{l_g,t}^{s,ini} \geq Pr_{l_g,t}^{s,ter} \quad (14)$$

$$f_{l_g,t}^{s2} = \gamma_{l_g} (Pr_{l_g,t}^{s,ini2} - Pr_{l_g,t}^{s,ter2}) \quad (15)$$

$$0 \leq f_{l_g,t}^s \leq f_{l_g,\max}^s \quad (16)$$

$$P_{gt,t}^s = c_{GT} f_{l_g,gt,t}^s \quad (17)$$

$$G_{n,t}^{s,hy} = \eta_e \frac{P_{n,t}^{s,P2G}}{\Omega_{hy}} \quad (18)$$

$$\begin{aligned} & \sum_{ig \in I_g} G_{ig,t}^s + \sum_{n \in N} G_{n,t}^{s,hy} + \sum_{l_g \in L_g} f_{l_g,t}^{s,ini} \\ & - \sum_{l_g \in L_g} f_{l_g,t}^{s,ter} = \sum_{k_g \in K_g} G_{k_g,t} + \sum_{l_g \in L_g} f_{l_g,gt,t}^s \end{aligned} \quad (19)$$

Constraints (20)–(29) model the water system. The output of water reservoir $\sigma_{wr,t}^s$ is limited in (20), where $\sigma_{wr,\max}$ is

the maximum limit. The water pressure limit for pipes with and without water pumps ($h_{w,t}^{s,lw}$ and $h_{w,t}^{s,lwp}$) is given in (21), where the minimum and maximum limits are $h_{w,\min}^{lw}$, $h_{w,\min}^{lwp}$, $h_{w,\max}^{lw}$, and $h_{w,\max}^{lwp}$. Similar to the Weymouth gas flow equation, Darcy-Weisbach equation is widely applied to describe water flow [43]. (22)–(26) are the hydraulic characteristics of water pipes based on the Darcy-Weisbach equation [44], where $\{\cdot\} = l_w, l_{wp}$ represent water pipes equipped with and without water pumps; $\tilde{h}_{l_w,t}^s$ and $\tilde{h}_{l_{wp},t}^s$ are the head loss of the pipes with and without water pumps; $h_{w,t}^{s,\{\cdot\},ini}$ and $h_{w,t}^{s,\{\cdot\},ter}$ are the pressure at the initial and terminal nodes while $\bar{h}_{w,t}^{s,\{\cdot\},ini}$ and $\bar{h}_{w,t}^{s,\{\cdot\},ini}$ represent the elevation of water nodes; $a_{l_{wp}}$, $b_{l_{wp}}$ are the water pump characteristic coefficients; R_{l_w} and $R_{l_{wp}}$ denotes the head loss coefficients; (22) and (24) determine the head loss. In (27), the power consumed by the water pumps is given, where $\sigma_{wp,t}^s$ is the power consumption and $f_{l_{wp},t}^s$ denotes the water flow. Constraint (28) limits the water flow in the water system. The balancing condition of the water system is presented in constraint (29), where $P_{n,t}^{s,P2G}$ and $\sigma_{k_w,t}$ are the P2G water consumption and node water demand.

Based on [27], the carbon emission modelling is given in equations (30)–(33). The carbon emission associated with the power and gas generation and water pumping is given in (30), denoted as Γ_{ie}^s , Γ_{ig}^s , and Γ_{wp}^s , respectively, where $c_{\{\cdot\}}$ is the carbon intensity coefficient. In (31), the carbon emission caused by power and gas flow is depicted. (32) and (33) define the emission index of power and gas systems.

$$0 \leq \sigma_{wr,t}^s \leq \sigma_{wr,\max} \quad (20)$$

$$h_{w,\min}^{\{\cdot\}} \leq h_{w,t}^{s,\{\cdot\}} \leq h_{w,\max}^{\{\cdot\}}, \quad \{\cdot\} = l_w, l_{wp} \quad (21)$$

$$\tilde{h}_{l_w,t}^s = \left(h_{w,t}^{s,lw,ini} + \bar{h}_{w,t}^{s,lw,ini} \right) - \left(h_{w,t}^{s,lw,ter} + \bar{h}_{w,t}^{s,lw,ter} \right) \quad (22)$$

$$\tilde{h}_{l_{wp},t}^s \geq 0 \quad (23)$$

$$\tilde{h}_{l_{wp},t}^s = \left(h_{w,t}^{s,lwp,ini} + \bar{h}_{w,t}^{s,lwp,ini} \right) - \left(h_{w,t}^{s,lwp,ter} + \bar{h}_{w,t}^{s,lwp,ter} \right) \quad (24)$$

$$\tilde{h}_{l_{wp},t}^s + a_{l_{wp}} f_{l_{wp},t}^s + b_{l_{wp}} = R_{l_{wp}} f_{l_{wp},t}^{s2} \quad (25)$$

$$\tilde{h}_{l_w,t}^s = R_{l_w} f_{l_w,t}^{s2} \quad (26)$$

$$\sigma_{wp,t}^s = \left(a_{l_{wp}} f_{l_{wp},t}^{s2} + b_{l_{wp}} f_{l_{wp},t}^s \right) / \pi_{wp} \quad (27)$$

$$0 \leq f_{\{\cdot\},t}^s \leq f_{\{\cdot\},\max}^s, \quad \{\cdot\} = l_w, l_{wp} \quad (28)$$

$$\begin{aligned} & \sum_{wr \in WR} \sigma_{wr,t}^s + \sum_{\{\cdot\} \in L_w, L_{wp}} f_{\{\cdot\},t}^{s,ini} - \sum_{\{\cdot\} \in L_w, L_{wp}} f_{\{\cdot\},t}^{s,ter} \\ & = \sum_{k_{pg} \in K_{pg}} \sigma_{k_{pg}} P_{n,t}^{s,P2G} + \sum_{k_w \in K_w} \sigma_{k_w,t} \end{aligned} \quad (29)$$

$$\Gamma_{\{\cdot\}}^s = c_{\{\cdot\}} P_{\{\cdot\},t}^s, \quad \{\cdot\} = i_e, i_g, wp \quad (30)$$

$$\Gamma_{\{\cdot\}}^s = c_{\{\cdot\}} f_{\{\cdot\},t}^s, \quad \{\cdot\} = l_e, l_g \quad (31)$$

$$CI_{e,t}^s = \frac{\sum_{ie \in I_e} P_{ie,t}^s c_{ie} + \sum_{le \in L_e} f_{le,t}^s c_{le}}{\sum_{ie \in I_e} P_{ie,t}^s + \sum_{j \in J} \omega_{j,t}^s + \sum_{le \in L_e} f_{le,t}^s} \quad (32)$$

$$CI_{g,t}^s = \frac{\sum_{ig \in I_g} P_{ig,t}^s c_{ig} + \sum_{lg \in L_g} f_{lg,t}^s c_{lg}}{\sum_{ig \in I_g} P_{ig,t}^s + \sum_{lg \in L_g} f_{lg,t}^s} \quad (33)$$

B. Piecewise Linear Approximation

The Weymouth gas flow equation shown in (15) is nonlinear and non-convex, yet Pr_t^2 is given in the squared form all the time, which does not require linearization. However, the nonlinear term $f_{lg,t}^{s2}$ can be linearized by piecewise linear approximation to convert the original problem into a Mixed-Integer Linear Programming problem [45], [46]. Meanwhile, $f_{l_{wp},t}^{s2}$ of (25)–(26) in the Darcy-Weisbach equation also requires the linearization. We use $h(f_t)$ represents the nonlinear function, Δf_k is the segment of gas pipelines and water pipes, and $\chi_{t,k}$ is the auxiliary continuous variable.

$$h(f_t) \approx h(\Delta f_1) + \sum_{k \in K} (h(\Delta f_{k+1}) - h(\Delta f_k)) \chi_{t,k} \quad (34)$$

$$f_t = \Delta f_1 + \sum_{k \in K} (\Delta f_{k+1} - \Delta f_k) \chi_{t,k} \quad (35)$$

$$0 \leq \chi_{t,k} \leq 1 \quad (36)$$

C. Real-Time Operation

A more precise renewable generation can be estimated after the first stage, which contributes to a more accurate and optimal decision making in the real-time stage. Constraints (37) and (38) are the adjusted generation limits and balancing condition of the power system, where $P_{\{\cdot\},t}^{re}$ is the adjusted output value in the second stage; $\xi_{j,t}^s$ is denoted as the uncertain renewable generation. Moreover, due to the space limitation, the rest of second-stage constraints, which are the same as the first-stage constraints, are omitted. The superscript ‘s’ on each variable is changed to ‘re’. ‘s’ represents the scheduled decision variables in the first stage and ‘re’ represents the adjusted decision variables in the second stage.

$$P_{\{\cdot\},t}^s - r_{\{\cdot\},t}^- \leq P_{\{\cdot\},t}^{re} \leq P_{\{\cdot\},t}^s + r_{\{\cdot\},t}^+, \quad \{\cdot\} = i_e, gt, wp \quad (37)$$

$$\begin{aligned} & \sum_{ie \in I_e} P_{ie,t}^s + \sum_{j \in J} \xi_{j,t}^s + \sum_{le \in L_e} f_{le,t}^{s,ini} - \sum_{le \in L_e} f_{le,t}^{s,ter} + P_{gt,t}^s \\ & = \sum_{k_e \in K_e} P_{k_e,t}^s + \sum_{le \in L_e} f_{le,t}^{s,inj} + \sum_{n \in N} P_{n,t}^{s,P2G} + \sum_{wp \in WP} P_{wp,t}^s \end{aligned} \quad (38)$$

D. Objective Function

The sub-objective functions of the two stages to be minimized are given in (39) and (40). Function (39) presents the day-ahead optimization objective including the generation cost, power purchase cost from the upper-level market, reserve preparation cost and carbon emission cost, where $\lambda_{\{\cdot\}}$ represents the cost

coefficients.

$$\begin{aligned} Obj_1 = \min & \sum_{ie \in I_e, ig \in I_g, t \in T, wr \in WR, gt \in GT} \lambda_m P_{m,t}^s \\ & + \lambda_{ie}^a P_{ie,t}^{s^2} + \lambda_{ie}^b P_{ie,t}^s + \lambda_{ie}^c \\ & + \lambda_{wr} P_{wr,t}^s + \lambda_{ig} P_{ig,t}^s + \lambda_c^e C I_{e,t}^s \\ & + \lambda_c^g C I_{g,t}^s + \lambda_{\{\cdot\}}^+ r_{\{\cdot\},t}^+ + \lambda_{\{\cdot\}}^- r_{\{\cdot\},t}^-, \{\cdot\} \\ & = i_e, gt, wp \end{aligned} \quad (39)$$

In (40), the redispatch and corrective actions are considered in the real-time problem, consisting of the penalty cost due to the overestimation or underestimation of the day-ahead generation dispatch, where $\lambda_{\{\cdot\}}^{re}$ is the adjustment cost coefficient.

$$\begin{aligned} Obj_2 = \min & \sum_{ie \in I_e, ig \in I_g, t \in T, wr \in WR} \lambda_m^{re} |P_{m,t}^s - P_{m,t}^{re}| \\ & + \lambda_{ie}^{re} |P_{ie,t}^s - P_{ie,t}^{re}| \\ & + \lambda_{ig}^{re} |P_{ig,t}^s - P_{ig,t}^{re}| \\ & + \lambda_{wr}^{re} |P_{wr,t}^s - P_{wr,t}^{re}| \end{aligned} \quad (40)$$

III. SOLUTION METHOD

In (41)–(44), a compact matrix form is adopted to represent the model (1)–(40). (41) and (42) are the second-stage recourse problem, where (41) is the second-stage objective function, determining the expected result under the worst-case distribution scenario. The first and second stage variables are represented by x and y , respectively. The uncertain variable is denoted as ξ . The second-stage constraints are represented by (42).

$$\mathcal{M}(x, \xi) = \sup_{\mathbb{P} \in P_0(\mathbb{R}^{|\mathcal{S}|})} E_{\mathbb{P}}[M(x, \xi)] \quad (41)$$

$$\text{Subject to } Tx + Wy + G\xi \leq h, \quad (42)$$

The overall model is given in (43)–(44), where (44) is the first-stage constraints.

$$\text{minimize}_{x \in X} c^T x + \mathcal{M}(x, \xi) \quad (43)$$

$$\text{Subject to } Ax \leq d, \quad (44)$$

A. Ambiguity Set and Semidefinite Formulation

The real uncertainty distribution of renewable generation might not be accurately estimated by SO. To this end, the ambiguity set of DRO is adopted to model all the possible distributions. (45) is the moment-based ambiguity set consisting of the known information of mean vector μ and covariance matrix Σ . $\mathbb{R}^{|\mathcal{S}|}$ is the set of all the uncertain variable ξ . $P_0(\mathbb{R}^{|\mathcal{S}|})$ denotes the set accommodating all the possible distributions.

$$\begin{aligned} \Omega(\mathcal{E}, \mu, \Sigma) &= \left\{ \mathbb{P} \in P_0(\mathbb{R}^{|\mathcal{S}|}) \mid \begin{array}{l} \xi \in \mathbb{R}^{|\mathcal{S}|} \\ P\{\xi \in \mathbb{R}^{|\mathcal{S}|}\} = 1 \\ E\{\xi\} = \mu \\ E[(\xi - \mu)(\xi - \mu)^T] \leq \Sigma \end{array} \right\} \end{aligned} \quad (45)$$

The explicit form of (45) can be shown in (46) and (47).

$$\mathcal{R}_P(x) = \max_{\mathbb{P}(\xi) \in \Omega(\mathcal{E}, \mu, \Sigma)} \int_{\mathbb{R}^{|\mathcal{S}|}} M(x, \xi) p_f(\xi) d\xi \quad (46)$$

Equation (46) is the second-stage objective function $\mathcal{M}(x, \xi)$ in explicit form, subject to constraints (47). And the decision variable is the probability density $p_f(\xi)$.

$$\begin{aligned} \Omega(\mathcal{E}, \mu, \Sigma) &= \left\{ \begin{array}{l} \int_{\mathbb{R}^{|\mathcal{S}|}} P(\xi) d\xi = 1 \\ \int_{\mathbb{R}^{|\mathcal{S}|}} \xi^j P(\xi) d\xi = \mu_j, j = 1, 2, \dots, \mathbb{R}^{|\mathcal{S}|} \\ \int_{\mathbb{R}^{|\mathcal{S}|}} \xi^j \xi^k P(\xi) d\xi = \Sigma_{jk} + \mu_j \mu_k, j, k = 1, \dots, \mathbb{R}^{|\mathcal{S}|} \end{array} \right\} \end{aligned} \quad (47)$$

To remove the inner maximization of (46), the following equivalent reformulation (48)–(49) is made based on the conic duality of (41), where δ_0 , ζ_j and χ_{jk} are the dual variables associated with constraints (47), Θ represents $\Sigma + \mu(\mu)^T$. Based on the strong duality theorem [47], $\mathcal{R}_P(x) = \mathcal{R}_D(x)$. The original infinite-dimensional linear program $E_{\mathbb{P}}[M(x, \xi)]$ and the primal form $\mathcal{R}_P(x)$ are transformed into (48)–(49) with a finite number of variables.

$$\mathcal{R}_D(x) = \min_{\chi, \zeta, \delta_0} \chi^T \Theta + \zeta^T \mu + \delta_0 \quad (48)$$

$$\text{Subject to } (\xi)^T \chi \xi + \zeta^T \xi + \delta_0 \geq \mathcal{M}(x, \xi), \forall \xi \in \mathbb{R}^{|\mathcal{S}|} \quad (49)$$

Note that the problem $\min_{x \in X} c^T x + \mathcal{R}_D(x)$ cannot be solved directly due to the infinite constraints of (49). Moreover, $\mathcal{M}(x, \xi)$ is not given in closed form. Therefore, the dual of $\mathcal{M}(x, \xi)$ is equivalently obtained as a positive quadratic function in closed form [48], where ϑ is the new dual variable.

$$\begin{aligned} (\xi)^T \chi \xi + (\psi + G^T \vartheta^i)' \xi + \delta_0 - (h - Tx) \vartheta^i &\geq 0 \\ \forall \xi \in \mathbb{R}^{|\mathcal{S}|}, i = 1, 2, \dots, N_v \end{aligned} \quad (50)$$

We use $\mathcal{F}$ as the polyhedral set incorporating all the extreme points.

$$\mathcal{F} = \{\vartheta \mid W^T \tau = q, \vartheta \leq 0\} \quad (51)$$

Equation (52)–(57) present the compact matrix form in semidefinite formulation. Thus, the final semidefinite form of (43) is given:

$$\min_{x, \chi, \zeta, \delta_0} c^T x + \chi^T \Theta + \zeta^T \mu + \delta_0 \quad (52)$$

$$\text{Subject to } x \in X \quad (53)$$

$$\begin{bmatrix} \xi \\ 1 \end{bmatrix}' \begin{bmatrix} M_{11} & M_{21} \\ M_{21}^T & M_{22} \end{bmatrix} \begin{bmatrix} \xi \\ 1 \end{bmatrix} \geq 0, \quad (54)$$

$$\forall \xi \in \mathbb{R}^{|\mathcal{S}|}, i = 1, 2, \dots, N_v, x \in X, \forall \vartheta^i \in \mathcal{F}$$

$$M_{11} = \chi \quad (55)$$

$$M_{21} = \frac{1}{2} (\zeta + G^T \vartheta^i) \quad (56)$$

$$M_{22} = \delta_0 - (h - Tx)^T \vartheta^i \quad (57)$$

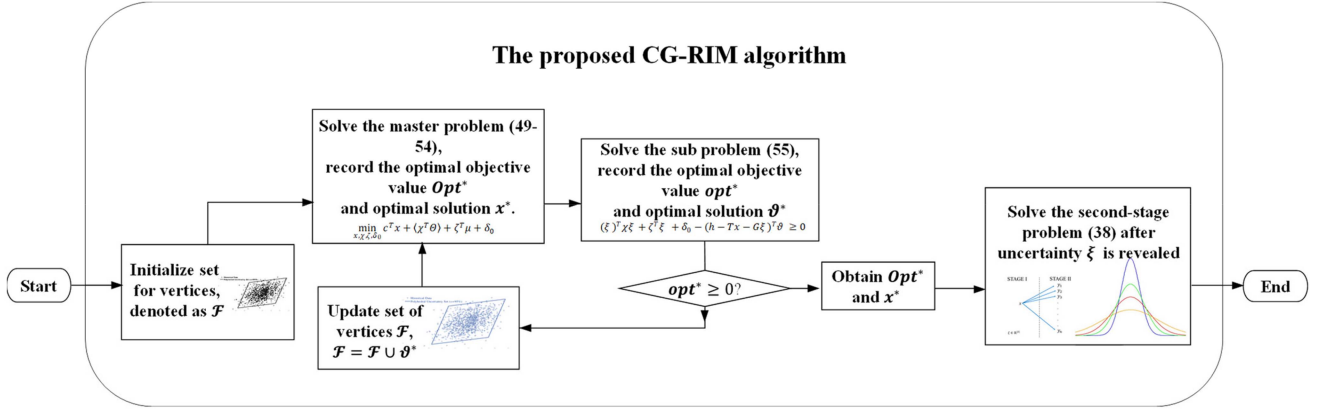


Fig. 2. Flowchart of the proposed CG-RIM.

 TABLE I
PARAMETERS OF WATER RESERVOIRS

Node No.	$P_{wr,max}$ (m ³ /h)	λ_{wr} (\$/m ³)	Elevation (m)
1	1325	6.4	-252.5
2	700	2.6	-255

 TABLE II
PARAMETERS OF GAS WELLS

Node No.	$P_{ig,max}$ (kcf/h)	$P_{ig,min}$ (kcf/h)	λ_{ig} (\$/kcf)
1	35.31	0	22
8	70.63	0	20

 TABLE III
PARAMETERS OF TRADITIONAL DGs

Bus No.	$P_{ie,max}$ (MW)	$P_{ie,min}$ (MW)	R_i^+, R_i^- (MW)	λ_{ie}^a (\$/MW ²)	λ_{ie}^b (\$/MW)	λ_{ie}^c (\$)
13	1.2	0.3	0.2	6000	7100	6200
23	1.2	0.3	0.2	4500	10500	4000
28	1.0	0.1	0.2	4500	10500	4000

 TABLE IV
COST COEFFICIENTS

Cost coefficient	λ_{wr} (\$/m ³)	λ_c^e (\$/t)	λ_c^g (\$/t)	$\lambda_{ie}^+, \lambda_{ie}^-$ (\$/MWh)	$\lambda_{gt}^+, \lambda_{gt}^-$ (\$/MWh)	$\lambda_{wp}^+, \lambda_{wp}^-$ (\$/MWh)
Price (\$)	2.5	50	40	60,60	50,50	0.08,0.08

B. Relaxation Iterative Method

Note that the problem (52)–(57) still has an exponential number of semidefinite constraints. Accordingly, instead of solving the global optimization with high computational burden, we implement a CG-RIM to enumerate part of all the vertices of the feasibility region in each iteration. The proposed CG-RIM enables to temporarily relax the majority of the semidefinite constraints. In Fig. 2, the CG-RIM solves the master and sub problems in each iteration, respectively.

$$\begin{aligned}
 &(\xi)^T \chi \xi + \zeta^T \xi + \delta_0 - (h - Tx - G\xi)^T \vartheta \geq 0 \\
 &\text{Subject to } \forall \xi \in \mathbb{R}^{|S|}, \tau \in \mathcal{F}
 \end{aligned} \quad (58)$$

IV. CASE STUDIES

A. Simulation Setup

The proposed two-stage DRO for LCM is analysed through simulation results on a modified IEEE 33-bus system, a 20-node gas system and a 10-node water system [49], [50], where generator information is given in Tables I, II, and III. All the tests are evaluated in daily scale. Please note that we use ‘kcf’ as the gas unit, representing 1000 cubic feet of natural gas [51]. Cost coefficients in (36) are given in Table IV [52], [12], [53]. The capacity of P2G is 2 MW and the conversion efficiency is 80%. The water consumption coefficient of P2G is 1.26 m³/MWh

 TABLE V
WATER PUMP CHARACTERISTICS

Node No.	a_{lwp}	b_{lwp}	π_{wp}
5	-1.1	7448	0.5

 TABLE VI
WATER PIPE COEFFICIENTS

Node No.	Diameter (m)	Length (m)	Head loss coefficient	Min pressure (m)	Max pressure (m)
1	0.25	40	0.039	-	-
2	0.25	130	0.126	-	-
3	0.3	370	0.299	40	80
4	0.25	100	0.097	40	80
5	0.3	430	0.323	35	80
6	0.25	260	0.252	35	80
7	0.25	80	0.078	40	80
8	0.25	500	0.484	40	80
9	0.25	350	0.339	40	80
10	0.25	130	0.126	40	80

[12]. The capacity of the PV generator is 0.6 MW. The carbon intensity coefficients of power and gas generation are $c_{ie} = 0.3$ t/MWh and $c_{ig} = 0.8$ t/MWh. The carbon intensity coefficient for the water pump is $c_{wp} = 0.363$ t/MWh [28], [54]. Note that the maximum water reservoir output $P_{wr,max}$ also represents the reservoir capacity [50]. Tables V and VI demonstrate the data of

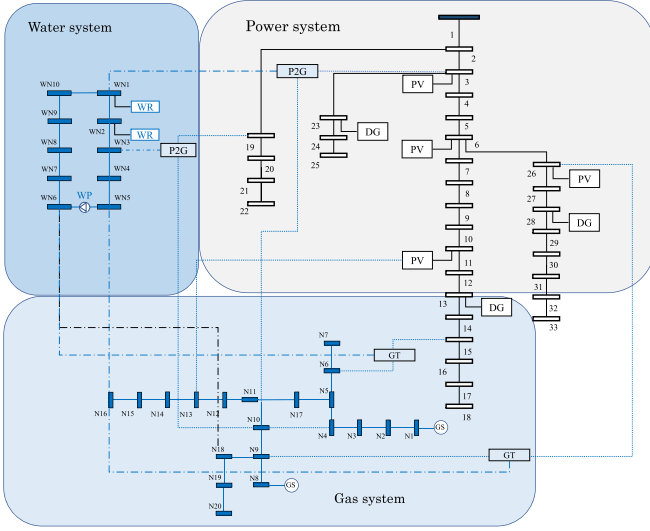


Fig. 3. Proposed test system under water-energy-carbon nexus.

water pump and water pipe. The power system has 3 traditional DGs and 4 renewable DGs. The power purchase from upper level market is \$7000/MWh [55]. Fig. 3 shows the system structure of LCWES with tight couplings among each subsystem. The load profile is given in Fig. 4. PV generators harness the solar energy and transform to electric power. And its generation profile is shown in Fig. 5. Traditional DGs and natural gas sources are applied for power and gas supply. Gas turbines are used to link the power and gas systems, transforming gas to power. We adopt P2Gs as the conversion technology from power to gas. In the water system, water reservoirs are the sources. Water pumps are used to pump and drain water. We define a notation for clarifying the infrastructures:

- PV Photovoltaic distributed generators.
- DG Traditional distributed generators.
- N Gas nodes.
- GS Natural gas sources.
- GT Gas turbines.
- P2G Power-to-gas converters.
- WN Water nodes.
- WR Water reservoirs.
- WP Water pumps.

The power and gas systems have 4 coupling points: i) buses 19 and 26 in the power system are connected with nodes 4 and 9 in the gas system via gas turbines, and ii) P2G facilities on buses 3 and 11 are connected with gas nodes 10 and 13. The gas pressure limits are given in Fig. 6. The water distribution system interconnects with both the power and gas systems: i) water nodes 1 and 3 are connected with the P2G facilities for the water electrolysis process, and ii) water pump at nodes 1, 2, and 5 consume power from the power system and converted power from the gas turbine. We solve the proposed LCM problem using MOSEK v10 in the MATLAB R2018b. All the simulations were

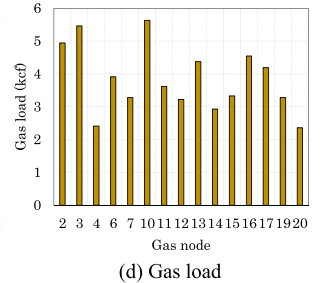
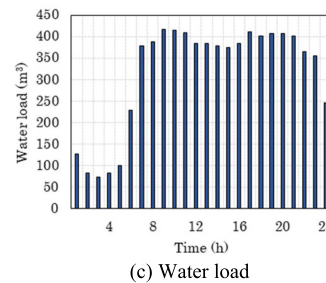
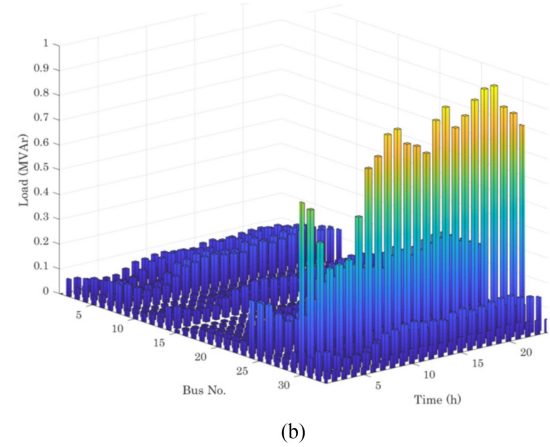
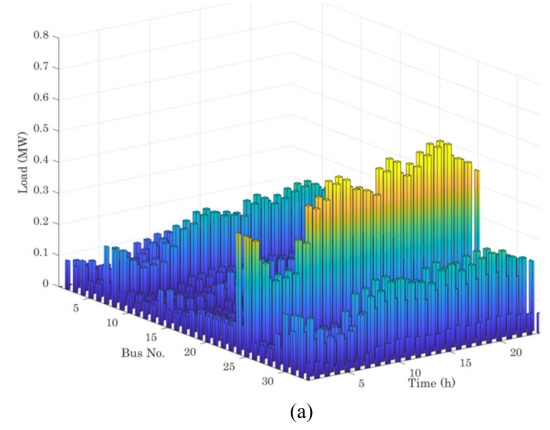


Fig. 4. Load demand profile.

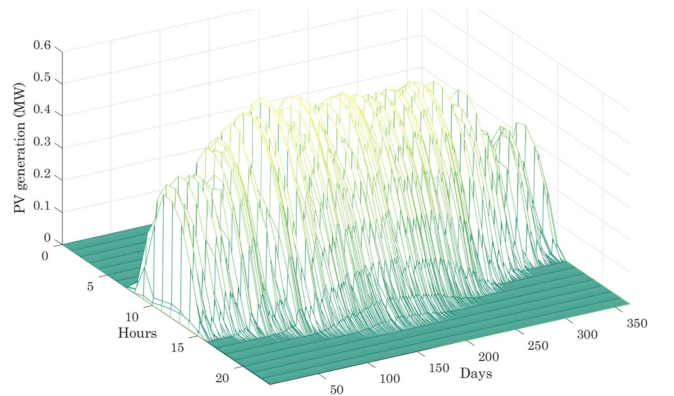


Fig. 5. PV generation profile.

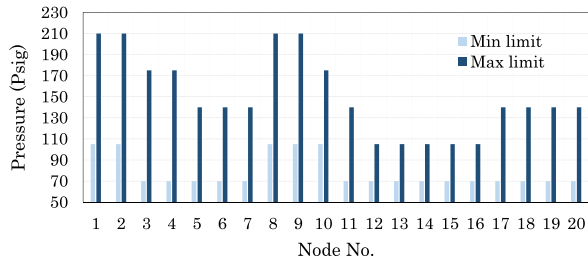


Fig. 6. Limits of gas pressure at each node.

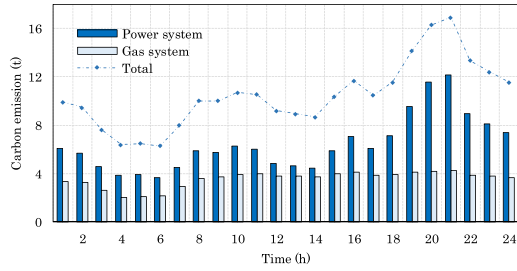


Fig. 7. Carbon emission curve of case 1.

performed on a personal computer with an Intel CPU (1.5 GHz) and a 16 GB RAM.

Our evaluation considers the five cases that feature different system parameters. Case 1 serves as a baseline case. Case 2 has a PV generator capacity twice that of baseline case, and Case 3 doubles the P2G facility capacity in Case 1. Case 4 considers a penalty cost of carbon emission twice that of Case 1, and Case 5 replaces all the traditional DGs with PV generators. Next, we analyse the proposed method's economic performance in different cases, and then report carbon emissions and water consumption.

B. Analysis of The Carbon Emission and Water Consumption

Fig. 7 depicts the carbon emission profile over a 24-hour window. As shown, high-carbon emissions occurred between 19:00 and 22:00. At the peak, total carbon emissions reached 16.9t at 21:00. Carbon emissions were observed at the lowest level between 4:00 and 6:00. The results show that carbon emissions by the power system is higher than that of the gas and water systems combined. Notably, the emissions by the water system accounts for 5.5% of the total emission only. At 21:00, the total carbon emission amounted to 16.9t and was comprised of 12.1t by the power system, 4.3t by the gas system, and 0.6t by the water system.

We examined the impact of a greater PV capacity on carbon emissions and water consumption by comparing single-stage robust optimization (SRO) with a box uncertainty set, two-stage robust optimization (TRO), and the proposed MDRO. As Fig. 8(a) shows, carbon emissions decrease with the PV capacity; SRO has the highest carbon emission and MDRO yields the lowest emission. With the PV capacity twice that of the baseline case, SRO and MDRO produce 233t and 210t emissions respectively.

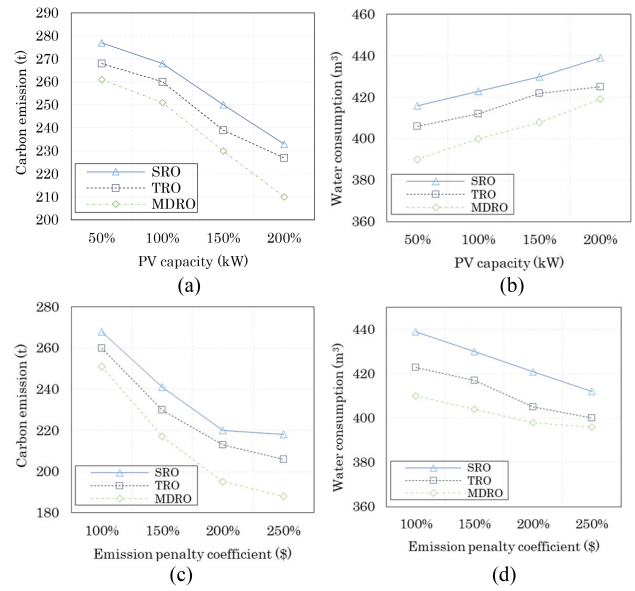


Fig. 8. (a). Impact of the PV capacity on carbon emission, (b). impact of the PV capacity on water consumption, (c). impact of the emission penalty coefficient on carbon emission, and (d). impact of emission penalty coefficient on water consumption.

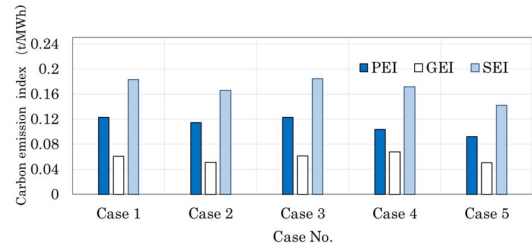


Fig. 9. Results of carbon emission indices.

The MDRO method shows a 20% holistic system carbon emission reduction when PV capacity increases from 50% to 200%. Fig. 8(b) indicates that water consumption increase with the PV capacity, because the equipped P2G facilities consume water too. The emission remains the lowest for MDRO, which models the uncertainty and variability in renewable power generation appropriately. Figs. 8(c) and (d) compare carbon emissions and water consumption when the emission penalty increases. In these scenarios, the optimization model tends to purchase more power from the upstream market, due to the higher emission penalty. Fig. 8(c) shows 25% of emission reduction when the penalty coefficient rises from 100% to 250%. Overall, in light of Fig. 8(a) and (c), carbon emissions are not significantly reduced. As indicated in (30) and (31), because apart from generation and emission penalty, consumption and distribution of water and energy are main reasons causing high emission amount.

Fig. 9 presents the results of emission indices, as formulated in equations (31) and (32), which include power system emission index (PEI), gas system emission index (GEI), and system emission index (SEI). These emission indices reveal the entire system's carbon intensity. Calculated as the sum of PEI and GEI, SEI reflects the carbon intensity of the entire system. In Case 1,

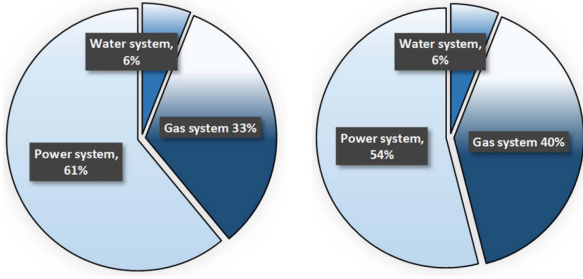


Fig. 10. The proportion of resulted carbon emission of cases 1 and 5.

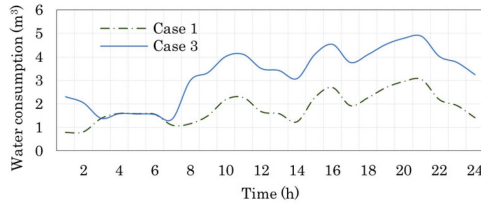


Fig. 11. Water consumption of P2G₁₉₋₄.

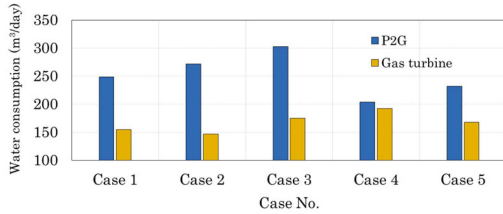


Fig. 12. Water consumption of P2G facilities and gas turbines.

PEI is 0.12 t/MWh, GEI is 0.06 t/MWh, and SEI is 0.18 t/MWh. Case 5 has the lowest PEI, GEI and SEI, suggesting that a higher penetration of renewable energy sources and decreased usage of traditional DGs can alleviate carbon emissions. Similar SEI values are observed in Cases 1 and 3, as well as Cases 2 and 4. Fig. 10 provides pie charts of carbon emission by the respective systems in Cases 1 and 5. As shown, the carbon emission by the power system in Case 1 accounts for the majority of the total emission (i.e., 61%). With all the traditional DGs replaced by renewable energy sources, the power system's portion of emission decreases by 7% and that of the gas system increases by 7%, while the emission by the water system remains the same (i.e., 6%). It can be observed that case 5 still shows a considerable proportion of carbon emissions. The reason is that apart from power generation, the power distribution still produces carbon emissions.

Figs. 11 and 12 indicate the water scheduling and total water consumption by energy conversion, respectively. Fig. 11 shows the scheduling of the P2G facility which interconnects between power bus 19 and gas node 4. Case 1 records 3 m³ water consumption at 20:00 whilst Case 3 has 5 m³ water consumption. Case 3 incurs a water consumption of approximately 81% more than that of Case 1, due to the doubled capacity of P2G facilities.

TABLE VII
ECONOMIC PERFORMANCE FOR ALL CASES

Economic result (\$)	Case 1	Case 2	Case 3	Case 4	Case 5
First-stage cost	943724	798742	939732	1171106	924260
Expected Second-stage cost	7580	8870	7420	7720	8430
Total cost	951304	807612	947152	1178826	932690

Fig. 12 presents the water consumed by all the P2G facilities and gas turbines. Overall, Case 3's water consumption is the highest, mainly because it allows an increased P2G conversion that in turn consumes more water. Noticeably, P2G exhibits greater water usage than that of gas turbines, implying a higher usage frequency of P2G in comparison with gas turbines. For instance, 249 m³ water is consumed by P2G in Case 1 but 155 m³ by gas turbines. When increasing the PV generator capacity (i.e., Case 3) by 100% from that in Case 2, water usage increases by 23 m³ whilst the gas turbines consume less water. That is, P2G facilities consume more water in Case 3 that features a PV capacity twice that in Case 2. Carbon emissions are prioritized in Case 4, so the water consumption by P2G facilities is significantly reduced, though the total water consumption only decreases by 4 m³. More gas turbine usage is scheduled as the result of a lower carbon emission..

C. Economic Performance of the Nexus System

Table VII summarizes the system operation cost in the respective cases. Overall, Case 2, which features a PV generator capacity twice that of Case 1, incurs the lowest operation cost, and Case 4 has the highest operation cost, due to the penalty for carbon emission is doubled. As a baseline, Case 1 has an operation cost of \$94,3724 in the day-ahead stage for system pre-dispatch, and \$7580 in the real-time response stage for adaptive recourse actions. The real-time redispatch cost in Case 2 is \$1290 higher than that of Case 1, because its greater PV capacity creates more variabilities in PV generation and therefore affects the real-time balance between supply and demand, according to the day-ahead operation scheme. Case 3 doubles the P2G conversion capacity in Case 1, so the additional power supplement can decrease the operation cost by consuming excessive PV generation. In Case 4, the carbon emission penalty is twice that of the baseline case, which leads to a 23% increase in the operation cost. However, the expected cost in the second stage remains mostly intact, because no carbon emission penalty is considered. In Case 5, when traditional DGs are replaced by PV generators, the carbon emission cost is assumed to be zero and results in a cost reduction of \$18614.

D. Scalability Analysis

An LCWES in a larger scale containing 123 buses, 40 gas nodes, and 20 water nodes (123–40–20) is further presented to verify the effectiveness of LCM. System topology and parameters are obtained from [49], [50], [56]. The LCWES contains seven 0.25 MW traditional DGs and six 0.3 MW PV generators.

TABLE VIII
BATTERY STORAGE PARAMETERS

Bus No.	Min power state (MWh)	Max power state (MWh)	Min dis/charging (MW)	Max dis/charging (MW)	Initial state (MWh)
74	0.02	0.09	0	0.04	0.04
83	0.03	0.14	0	0.06	0.06
95	0.02	0.09	0	0.04	0.05
103	0.03	0.14	0	0.06	0.08

TABLE IX
GAS STORAGE PARAMETERS

Node No.	Min gas state (kcf)	Max gas state (kcf)	Min dis/charging (kcf/h)	Max dis/charging (kcf/h)	Initial state (kcf)
5	10	60	0	10	45
11	10	100	0	10	60
25	10	60	0	10	45
31	10	100	0	10	60

TABLE X
ECONOMIC PERFORMANCE FOR ALL CASES

Economic result (\$)	Case 1	Case 2	Case 3
Power system	861248	820192	850965
Gas system	43215	43421	44180
Water system	152106	152122	152807
Carbon emissions	20324	17034	22098

TABLE XI
CARBON EMISSION BREAKDOWN FOR ALL CASES

Carbon emission proportion (%)	Case 1	Case 2	Case 3
Power system	45%	42%	44%
Gas system	35%	36%	35%
Water system	20%	22%	21%

In addition to sections A-E tested in the 33–20–10 system, we further incorporate the carbon emissions from the upper-level market to ensure the carbon emissions are effectively reduced. We also consider storage devices in this test environment. The distributed battery and gas storage information are given in Tables VIII and IX. The water storage tanks are located at nodes 5 and 8 with 100 m³ capacity. The following 3 cases are studied to test the performance of LCWES in a larger-scale system.

Case 1: Baseline case.

Case 2: Applying 120% the capacity of PV generators.

Case 3: Considering 120% penalty cost of carbon emissions.

We demonstrate the operation cost results of each sector in Table X. Due to the 20% additional PV capacity installed, it can be observed that Case 2 shows lower operation costs of the power system and carbon emissions than those of Case 1. In Case 3, the 20% higher penalty cost of carbon emissions causes \$37281 additional total operation cost. Table XI shows the source of carbon emissions. The power system carbon emission ranges

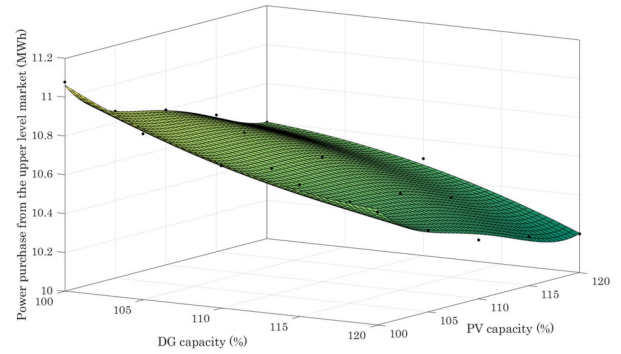


Fig. 13. Power purchase amount from the upper level market.

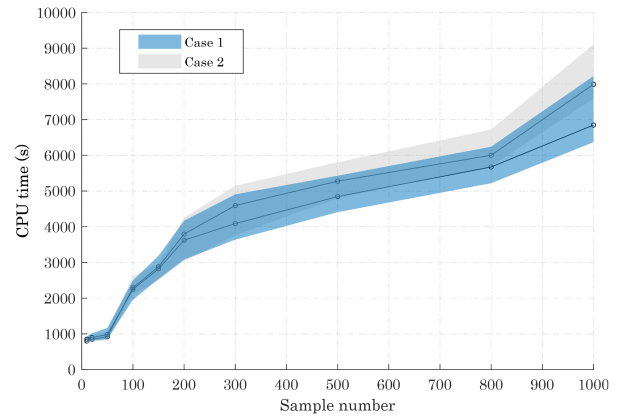


Fig. 14. Computational time of cases 1 and 2.

between 42% and 44% whilst the water system carbon emission varies between 20% and 22%. Fig. 13 presents the power purchase amount from the upper-level market. The capacity of the traditional DGs and PV generators ranges between 100% and 120%. The market power purchase decreases when the capacity increases. At the original system setting, the market power purchase is 11.08 MWh. When the DG and PV capacities are at 120%, the market power purchase is 10.20 MWh. Fig. 14 shows the computational performance of the proposed LCM problem tested in the 123–40–20 LCWES. We test each case 10 times and the results demonstrate the time range. Overall, we can observe that the computational time of Case 2 is higher than that of Case 1. When the sample number is 1000, the computational time of Cases 1 and 2 are up to 8217 seconds 9107 seconds.

V. CONCLUDING REMARKS

This paper develops a novel two-stage LCM model for the water-energy-carbon nexus in an integrated system to produce an optimal joint operation scheme. Specifically, our operation model incorporates essential sustainability issues that pertain to carbon emissions and water waste, and considers the close linkages of power, gas and water systems to enable economic operations of an integrated LCWES. A two-stage moment-based DRO approach is employed to make corrective decisions, according to real-time renewable estimations, which mitigates excessively conservative and suboptimal operations. We conduct

case studies to examine the proposed method's economic (operation) efficiency, water consumption and carbon emissions, compared with several prevalent benchmark methods. These cases feature different asset capacities and system parameters. The empirical results indicate that the proposed MDRO decreases carbon emission by 4.3% and water consumption by 2.9%, in comparison with TRO. Its two-stage setting further contributes to a 3.1% reduction in carbon emission and a 2.5% saving in water consumption than SRO. Our method can generate a practical, optimal operation scheme for system operators to maintain an economically efficient, sustainable, and low-carbon economy. It considers the linkages in the water-energy-carbon nexus and is capable of addressing the uncertainty and variability in renewable power generation. The resulting optimal two-stage flexible management scheme explores the tradeoff between economic objective and carbon emissions to support reliable and economic operations in urban energy systems, with reduced water waste and carbon emissions.

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